# Day 5 - Oct 24: Loops, GST, Sort/Filter, COUNTIFS, IF, IFERROR

Swagat | Revision sheet | Open: SQL 5-question check (Oct 25 afternoon)

## 1. KEY IDEAS

### Python: Conditions recap (re-read, recall, revise)

- if block = colon + 4-space indent. elif is checked only if every test above was False; strictest condition first. not, then and, then or.
- A chained comparison such as 60 <= 72 < 80 returns True or False, never the middle number. It means 60 <= 72 and 72 < 80.
- Leap year order: 400, then 100, then 4, else not. Tested 1900, 2000, 2024, 2023.

### Python: Loops

- for i in range(...): runs the block once per value. range(stop) starts at 0 and never includes stop. range(start, stop, step). range(10, 1) is empty (no error). Step 0 is a ValueError.
- Sum pattern: total = 0 BEFORE the loop; total = total + i inside. total = 0 inside the loop resets it every pass.
- Count pattern: count = 0 first, count = count + 1 inside an if. Max pattern: highest = marks[0], then if m > highest: highest = m. Never start from 0 (all-negative lists).
- while needs three parts: start, test, update. Missing update = infinite loop. The test runs first, so the body can run zero times.
- break leaves the innermost loop for good. continue skips the rest of ONE pass. In a while loop, continue jumps over the update line if the update sits after it: infinite loop. In a for loop it is safe, because range supplies the next value.
- f-string: only what is inside { } is calculated. Sum of 1 to N also equals n * (n + 1) // 2.

### Excel

- GST: =A2*$D$1. The rate is locked, the price moves. Store the rate as 18%, not 18. The header should say GST amount if the column holds only the tax.
- Grid: =$A2*B$1. The $ goes in front of the part that STAYS. Ask: when I drag, does this part stay or change? On a phone drag right first, then down with the row selected, or copy and paste over the block.
- Sorting: select the whole table with headers, tick My data has headers, so rows stay together. Sorting one column alone scrambles the rows.
- COUNT counts numbers only: it skips empty cells, text, numbers stored as text, TRUE/FALSE and errors. It also counts rows hidden by a filter.
- SUBTOTAL(102, range) counts visible rows only. 101 AVERAGE, 103 COUNTA, 104 MAX, 105 MIN, 109 SUM.
- COUNTIFS(range1, cond1, range2, cond2): all pairs must pass (AND). The text must match the cell exactly (Phone is not Mobile). No sum range, unlike SUMIFS.
- IF(test, if true, if false). A number in quotes is text, so 50000 in quotes breaks the test. AND and OR return ONE TRUE/FALSE, so give them single cells and drag down. Whole ranges mislead without any error.
- IFERROR(formula, "No data") hides every error, including real bugs. IFNA hides only #N/A.

## 2. EXAMPLES

```
total = 0
for i in range(1, n + 1):
    total = total + i
 print(total)
```

```
highest = marks[0]
for m in marks:
    if m > highest:
        highest = m
```

```
for i in range(1, 11):
    print(f"{num} x {i} = {num*i}")
```

```
=A2*$D$1                        GST amount
=$A2*B$1                        price x rate grid
=SUBTOTAL(102,D2:D9)            visible rows only
=COUNTIFS(A2:A9,"North",B2:B9,"Phone")
=IF(AND(A2="North",D2>=50000),"Star","-")
=IF(OR(A2="West",C2>=5),"Check","-")
=IFERROR(D7/(C7-1),"No data")
```

## 3. MY MISTAKES AND THE FIX

- 60 <= 72 < 80: answered 72 -> a comparison returns True/False. I knew the logic, but check the output type.
- Max pattern: started with 90 and tested m == highest -> start from marks[0] and test m > highest. Starting from 0 prints 0 for [-5,-2,-9], not None.
- while with no update: said it prints 1, 2, 3 and wrote 'while = 0' -> it prints 1 forever; while is a keyword; add n = n + 1.
- continue in a while loop: said 0,1,3,4 -> it hangs after 1, because continue skips the update. Also traced the wrong version of the code: always trace the exact code on screen.
- Sum of 1 to N: wrote range(1, n) -> range(1, n + 1). Multiplication table: wrote num*i outside the braces, so it printed num*i as text. Both fixed with Gemini's hint; rewrote both alone from a blank cell.
- My own conditions example: forgot to state the output; its else also catches games I never listed. Mixed wording in print messages; 1-space indent (use 4).
- GST: typed A21 for B2 (misread which row is first); forgot the = twice.
- Grid: put the $ on the part that moves, the reverse of the rule -> $ goes on the part that stays. Then typed $A1 instead of $A2. Then D1 held 8% instead of 18%, so D5 showed 32: working backwards (32 / 400) found it.
- Said COUNT counts empty cells -> it skips them. Filtered COUNT gives 8 because it ignores the filter.
- COUNTIFS gave 0: typed Mobile, the data said Phone -> exact spelling matters. Named rows 6 and 8 instead of 6 and 9 -> check Region AND Product in each row.

- IF: used A2 instead of D2, and put 50000 in quotes. AND over A2:A9 and D2:D9 collapsed to one answer for the whole column -> use A2 and D2, then drag.

## 4. HOW I IMPROVED

- Recall on Conditions was complete in my own words, and I fixed the 'a vs A / missing colon' type of slip without help.
- Excel: from guessing the $ to building =$A2*B$1 from four stay/change answers, and finding the 8% typo by working backwards.
- Loops: gave the output AND the reason for range, break and the sum/count patterns.

## 5. PRACTICE TO RETRY (no solutions)

- Sum of 1 to N (test 0, 1, 5, 100), multiplication table for 5 and 12, highest of [-5, -2, -9], while countdown 10 to 8, while with continue and a safe update.
- Excel: GST on 20 prices, price x rate grid from a blank sheet, SUBTOTAL vs COUNT on a filter, COUNTIFS with two conditions, IF with AND, IF with OR, IFERROR on a divide by zero.

## 6. RECALL QUESTIONS (answer without notes)

- What are the first and last values of range(2, 11, 3)? Why is the stop value left out?
- Why must total = 0 sit before the loop? What does it print if it sits inside?
- Why does highest = 0 fail on [-5, -2, -9]?
- Why does continue hang a while loop but not a for loop?
- Which part gets the $ in =$A2*B$1, and what does each $ stop?
- COUNT vs SUBTOTAL(102) on a filtered sheet. What does COUNT skip?
- Why does AND(A2:A9="North", D2:D9>=50000) mislead? How do you fix it?

## 7. OPEN ITEMS

- SQL 5-question check (closes Oct 22): Oct 25 afternoon. Day 5 README commit to learning-log: pending.